

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1727

Tricou 138 kV

April 18, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1727. The Interconnection Customer proposes a 62.9 MW solar photovoltaic facility interconnecting at the Tricou 138 kV substation in the Virginia area. The request was received on April 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 605 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,212,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                          | Contingency                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------|-----------------------------------------------|-------------|--------------|----------|
| DUPLESSIS-HYAK 138 KV                       | CEMENT PLANT-43RD & S. LAND 138 KV            | 136.4       | 243          | 25.8     |
| US ORD DEPOT-HERLONG-DUPLESSIS 138 KV       | CEMENT PLANT-43RD & S. LAND SUBSTATION 138 KV | 130.3       | 249          | 25.8     |
| CEMENT PLANT-43RD & S. LAND 138 KV          | TOLSON-US ORD DEPOT-HERLONG 138 KV            | 127.1       | 366          | 27.3     |
| VIRGINIA TECHPOWER PLANT-MIDWAY 138 KV Line | E PORTAL-PINNACLE PEAK SRP 138 KV             | 126.1       | 218          | 24.6     |
| OROVADA-TOLSON 138 KV                       | 43RD & S. LAND-CEMENT PLANT 138 KV            | 118.6       | 353          | 25.8     |
| US ORD DEPOT-HERLONG-TOLSON 138 KV          | CEMENT PLNT-43RD & S. LAND 138 KV             | 117.4       | 367          | 25.8     |
| 43RD & S. LAND 345/138kV Autotransformer    | TOLSON-US ORD DEPOT-HERLONG 138 KV            | 115.9       | 401          | 27.3     |
| NASSAU-LITCHFIELD 138 KV L                  | LOST NATION-NO TROY 138 KV                    | 114.2       | 192          | 28.3     |
| SPRINGBANK-LITCHFIELD 138 KV                | LOST NATION-NO TROY 138 KV line               | 112.8       | 214          | 28.3     |
| S CHRISTIANSBURG-MIDWAY 138 KV              | E PORTAL-PINNACLE PEAK SRP 138 KV Line        | 111.6       | 281          | 24.6     |
| SOUTH CHRISTIANSBURG-HAPPY CANYON 138 KV L  | PINNACLE PEAK SRP-E PORAL 138 KV              | 110.7       | 309          | 24.6     |
| HAPPY CANYON-EDGEMONT 138 KV                | PINNACLE PEAK SRP-EAST PORTAL 138 KV          | 109.8       | 336          | 24.6     |
| EDGEMONT-SHULLSBURG 138 KV line             | EAST PORTAL-PINNACLE PEAK SRP 138 KV          | 109.2       | 363          | 24.6     |

| Monitored Facility                                  | Contingency                       | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------------|-----------------------------------|-------------|--------------|----------|
| BEAR-MECKLENBURG POWER STA 345 KV                   | BEAR-CLOVER 345 KV                | 105.2       | 3,288        | 5.2      |
| BLUEFIELD AVENUE-PIUTE 138 KV                       | LOST NATION-NO TROY 138 KV        | 103.3       | 198          | 28.3     |
| VIRGINIA TECH POWER PLANT-HARTMAN CREEK 138 KV Line | PINNACLE PEAK SRP-E PORTAL 138 KV | 102.0       | 240          | 24.6     |
| BEAR-LEAVITT 345 KV                                 | LEAVITT-BEAR 345 KV               | 100.9       | 2,652        | 5.2      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                      | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| HYAK-DUPLESSIS 138 KV                                         | Rebuild line with higher-capacity conductor | \$2,339,000   | 4% (\$86,000)         |
| US ORD DEPOT-HERLONG-DUPLESSIS 138 KV                         | Rebuild line with higher-capacity conductor | \$7,326,000   | 4% (\$270,000)        |
| CEMENT PANT-43RD & S. LAND 138 KV                             | Rebuild line with higher-capacity conductor | \$12,684,000  | 3% (\$431,000)        |
| VIRGINIA TECH POWER PLANT SUBSTATION-MIDWAY SUBSTATION 138 KV | Rebuild line with higher-capacity conductor | \$4,699,000   | 5% (\$236,000)        |
| OROVADA-TOLSON 138 KV                                         | Rebuild line with higher-capacity conductor | \$13,638,000  | 3% (\$446,000)        |
| US ORD DEPOT-HERLONG-TOLSON 138 KV                            | Rebuild line with higher-capacity conductor | \$11,537,000  | 3% (\$377,000)        |
| 43RD & S. LAND 345/138kV Autotransformer                      | Replace transformer with higher-rated unit  | \$15,064,000  | 3% (\$512,000)        |
| NASSAU-LITCHFIELD 138 KV L                                    | Rebuild line with higher-capacity conductor | \$3,955,000   | 9% (\$352,000)        |
| SPRINGBANK SUBSTATION-LITCHFIELD SUBSTATION 138 KV            | Rebuild line with higher-capacity conductor | \$5,426,000   | 9% (\$483,000)        |
| S CHRISTIANSBURG-MIDWAY 138 KV                                | Rebuild line with higher-capacity conductor | \$8,105,000   | 5% (\$407,000)        |
| SOUTH CHRISTIANSBURG-HAPPY CANYON 138 KV L                    | Rebuild line with higher-capacity conductor | \$3,314,000   | 5% (\$167,000)        |
| HAPPY CANYON-EDGEMONT 138 KV line                             | Reconductor line                            | \$339,000     | 5% (\$17,000)         |
| EDGEMONT-SHULLSBURG 138 KV                                    | Reconductor line                            | \$1,164,000   | 5% (\$59,000)         |
| BEAR-MECKLENBURG POWER STA 345 KV                             | Reconductor line                            | \$68,915,000  | 0% (\$360,000)        |
| PIUTE-BLUEFIELD AVENUE 138 KV                                 | Reconductor line                            | \$4,981,000   | 9% (\$443,000)        |

| Facility                                        | Required Upgrade           | Cost Estimate | Allocation to Project |
|-------------------------------------------------|----------------------------|---------------|-----------------------|
| HARTMAN CRK-VIRGINIA TECH<br>POWER PLANT 138 KV | Reconductor line           | \$1,417,000   | 5% (\$71,000)         |
| BEAR-LEAVITT 345 KV Cir. 1                      | Reconductor line           | \$75,443,000  | 1% (\$495,000)        |
|                                                 | Total allocated to project |               | \$5,212,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.